



COMP90050: Advanced Database Systems

Lecturer: Farhana Choudhury (PhD)

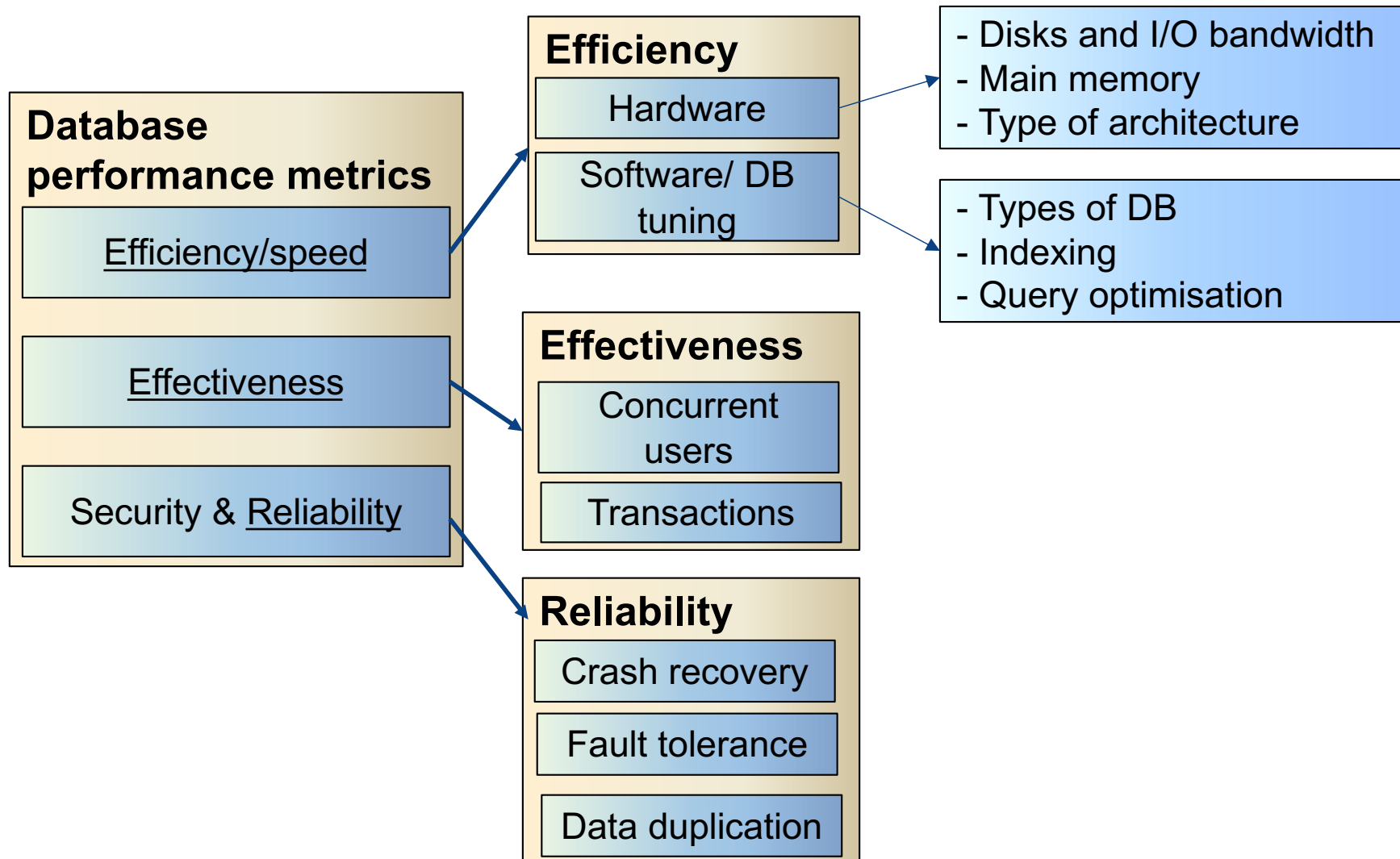
Database types

Week 1





Core Concepts of Database management system





Types of Database Systems

How the data are stored –

Simple file

- As a plain text file. Each line holds one record, with fields separated by delimiters (e.g., commas or tabs)

RDBS

- As a collection of tables (relations) consisting of rows and column. A primary key is used to uniquely identify each row.

Object oriented

- Data stored in the form of 'objects' directly (like OOP)

No-SQL

- Non relational – database modelled other than the tabular relations. Covers a wide range of database types.



Simple file

A terminal window with a grey title bar containing three colored circles (red, yellow, green). The window displays a simple file with three lines of text: '1, John Doe', '2, Jane Li', and '3, Rain Smith'. A blue cursor is positioned at the end of the third line.

```
1, John Doe
2, Jane Li
3, Rain Smith
```

- Can be fast for very simple applications
- Slow/unapplicable for complex applications
- Very hard to maintain (**concurrency** problems)
- Application dependent optimisation - many of the required features (that exist in relational databases) need to be incorporated manually. Unnecessary code development and potential increase in unreliability

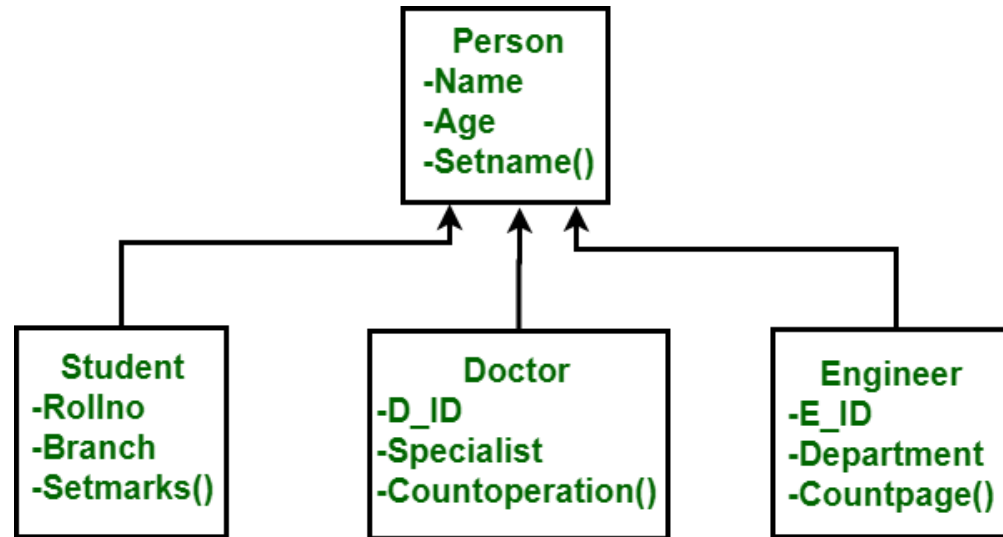
Relational DB systems

Students Table		Activities Table		
Student	ID*	ID*	Activity*	Cost
John Smith	084	084	Swimming	\$17
Jane Bloggs	100	084	Tennis	\$36
John Smith	182	100	Squash	\$40
Mark Antony	219	100	Swimming	\$17
		182	Tennis	\$36
		219	Golf	\$47
		219	Swimming	\$15
		219	Squash	\$40

Source: <http://www.databasedev.co.uk>

- Most commonly used, well suited to many applications
 - *Very reliable* (consistency of data – we will learn more later)
 - Application independent optimisation, very fast in most cases
- Can be slow for some simple applications

Object Oriented (OO) DB Systems



Source:<https://www.geeksforgeeks.org/basic-object-oriented-data-model/>

- Stores as objects (data and methods) directly, not tables
- Easier for applications running on OOPs
- Limited applications. Many commercial systems started did not survive the force of RDB technology.



NoSQL (also called Not Only SQL)

- Flexible/ no fixed schema (unlike RDB)
- Provides a mechanism for storage and retrieval of data modelled in means other than the tabular relations
- Simple design, should linearly scale

```
db.students.insert( {name: "Jack", born: 1992} )// add a doc
db.students.insert( {name: "Jill", born: 1990} )// add a doc
db.students.find( {name: "Jill"} )// list all docs where name field = 'Jill'
db.students.update( {name: "Jack"}, {$set: {born: 1990}} )
db.students.remove( {born: 1990} ) // delete docs where year = 1990
```

```
db.students.insert( {name: "John", profession: "engineer"} )
// add a new student – different schema but still works
```



NoSQL (also called Not Only SQL)

- NoSQL has ***compromise consistency*** and allows replications - We will discuss this more later.
- Most NoSQL databases offer "***eventual consistency***", which might result in reading data from an older version, a problem known as stale reads – we will learn more later

Types of NoSQL databases

Key Value



Example:
Riak, Tokyo Cabinet, Redis
server, Memcached,
Scalaris

Document-Based



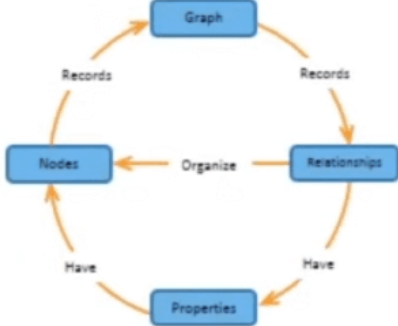
Example:
MongoDB, CouchDB,
OrientDB, RavenDB

Column-Based



Example:
BigTable, Cassandra,
Hbase,
Hypertable

Graph-Based



Example:
Neo4J, InfoGrid, Infinite
Graph, Flock DB

Image source: <https://jameskle.com/writes/no-sql>



Key-value pair database systems

Stores data as a collection of key–value pairs, each key is unique

Why useful - Many applications do not require the expressive functionality of transaction processing

- Used for very fast, highly parallel processing of large data
- Highly scalable
- Easy to move from one system to another

Drawbacks

- Atomic updates at Key-value pair level (row update) only.
- Simpler query



Some other DB systems – Deductive database systems (DDBS)

- Allows recursion
- There are no commercially available systems like RDBs
- Many applications do not require the expressive power of these systems (e.g. many commerce related applications)
- Many RDBs do provide some of the functionality – e.g. supporting transitive closure operation (a form of recursion in SQL2).
- E.g.,

`path(X,Y) :- edge(X,Y).`

`path(X,Y) :- edge(X,Z), path(Z,Y).`



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